

SHAPAI, Kazakhstan

WMO: 353020

Lat: 50.200N

Lon: 51.167E

Elev: 56

StdP: 14.67

Time Zone: 5.00 (E05)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-10.1	-21.2	1.7	-14.7	-15.8	2.3	-9.6	25.9	19.5	22.8	19.6	4.4	50	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	98.0	68.1	94.9	67.0	91.5	65.9	70.8	89.7	69.1	88.6	67.7	87.2	8.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	92.0	77.9	62.6	85.1	76.3	60.5	78.8	75.8	34.6	89.6	33.2	88.3	32.0	87.4	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.0	19.7	17.6	DB	-20.1	102.4	6.7	2.9	-24.9	104.5	-28.9	106.2	-32.7	107.8	-37.5	109.9	(4)
			WB	-19.8	73.3	6.5	3.0	-24.6	75.5	-28.4	77.2	-32.0	78.9	-36.8	81.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	14.0	15.3	29.6	49.1	62.5	71.7	76.4	74.2	60.7	45.7	30.8	20.0	(6)
		DBStd	24.37	12.79	12.45	10.33	8.85	8.63	8.31	6.26	7.68	8.15	8.41	10.77	11.58	(7)
		HDD50	4553	1116	971	634	120	8	0	0	11	188	575	930		(8)
		HDD65	7945	1581	1391	1098	480	154	31	2	15	175	598	1025	1395	(9)
		CDD50	3097	0	0	1	93	396	650	818	749	333	56	1	0	(10)
		CDD65	1013	0	0	0	3	77	230	355	300	47	1	0	0	(11)
		CDH74	12619	0	0	0	52	951	2892	4345	3718	646	15	0	0	(12)
		CDH80	6022	0	0	0	6	352	1401	2152	1878	230	3	0	0	(13)
(14)	Wind	WSAvg	7.6	8.6	8.9	8.6	8.3	7.4	6.8	6.5	6.5	6.3	7.1	7.6	8.1	(14)
(15)	Precipitation	PrecAvg	10.2	0.7	0.6	0.7	0.9	0.9	1.3	0.8	0.7	0.8	1.1	0.9	0.7	(15)
(16)		PrecMax	17.3	1.9	2.1	1.6	3.3	2.9	3.4	2.9	4.0	3.3	2.4	2.0	1.8	(16)
(17)		PrecMin	5.2	0.3	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.2		(17)
(18)		PrecStd	3.2	0.4	0.4	0.5	0.7	0.6	0.9	0.7	0.9	0.9	0.6	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	37.8	37.8	60.3	80.3	92.1	100.7	101.9	101.6	93.2	75.4	56.4	42.5		(19)
(20)		MCWB	36.3	35.0	49.0	59.7	64.0	69.6	68.9	68.6	64.1	56.8	50.0	39.8		(20)
(21)		DB	34.8	35.2	52.6	74.4	88.0	96.7	97.3	97.7	86.2	70.2	50.4	38.1		(21)
(22)		MCWB	33.4	33.3	44.2	57.2	63.6	67.8	67.7	67.9	62.0	54.7	45.7	35.9		(22)
(23)		DB	33.4	33.7	47.0	69.9	84.3	92.3	94.3	94.1	81.3	64.9	46.9	35.0		(23)
(24)		MCWB	32.2	32.2	41.4	55.4	62.6	65.9	67.0	66.8	60.5	52.2	43.5	33.2		(24)
(25)		DB	31.7	32.3	42.4	65.2	79.4	88.4	90.8	90.3	76.6	60.0	43.8	33.4		(25)
(26)		MCWB	30.7	31.0	37.7	52.6	60.7	65.3	66.4	65.8	58.8	49.9	41.1	31.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	36.8	35.8	49.6	61.3	67.2	77.1	72.6	72.8	66.9	60.5	51.8	41.0		(27)
(28)		MCDB	37.3	37.0	57.8	75.8	84.4	93.5	91.3	91.2	86.1	70.1	55.7	42.6		(28)
(29)		WB	33.7	33.8	45.2	58.5	65.1	70.9	70.5	69.9	63.9	56.3	46.9	36.2		(29)
(30)		MCDB	34.5	34.7	52.0	72.0	83.5	88.1	89.9	90.4	81.0	66.6	49.5	37.7		(30)
(31)		WB	32.3	32.3	41.6	56.3	63.4	67.9	69.1	68.4	61.8	53.5	43.9	33.6		(31)
(32)		MCDB	33.3	33.5	46.6	68.3	81.2	87.6	88.1	89.5	77.7	62.6	46.6	34.7		(32)
(33)		WB	30.6	31.0	38.3	53.7	61.5	65.9	67.8	67.0	59.8	50.9	41.2	32.1		(33)
(34)		MCDB	31.8	32.3	42.0	63.9	77.8	85.2	86.5	87.4	74.5	58.7	43.7	33.5		(34)
(35)	Mean Daily Temperature Range	MDBR	12.3	13.7	14.0	20.4	21.7	22.9	22.5	23.9	23.3	18.2	11.4	10.4		(35)
(36)		5% DB	8.2	10.2	20.3	26.3	27.0	27.8	26.7	28.5	30.7	26.9	15.0	8.1		(36)
(37)		MCWBR	8.1	9.4	13.1	12.9	10.1	9.2	8.4	9.5	13.1	15.0	11.0	7.2		(37)
(38)		5% WB	8.8	10.1	18.6	24.6	25.1	24.5	23.4	25.7	27.1	24.5	13.4	7.3		(38)
(39)	Clear-Sky Solar Irradiance	taub	0.321	0.342	0.394	0.446	0.425	0.412	0.416	0.428	0.382	0.362	0.340	0.319		(40)
(41)		taud	2.063	2.017	2.055	2.050	2.191	2.296	2.292	2.229	2.319	2.298	2.241	2.128		(41)
(42)		Ebn at Noon	211	241	251	252	264	269	266	255	255	236	208	194		(42)
(43)		Edh at Noon	27	37	43	48	44	40	39	40	33	28	23	22		(43)
(44)	All-Sky Solar Radiation	RadAvg	345	706	1070	1472	1925	2106	2038	1758	1254	718	331	229		(44)
(45)		RadStd	85	101	162	111	138	165	127	111	122	87	42	55		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page